



Principal Interests

Environmental management, Climate change Adaptation and Mitigation, and Disaster risk reduction | Application of Geographic Information Systems (GIS) and Remote Sensing (RS) | Geospatial Machine Learning

Academic Background

16 November 2025 Bachelor of Science (B.S.), Department of Geography and Environment, University of Dhaka; CGPA – 3.74 out of 4.00

December 2026 (Expected) Master of Science (M.S) (Thesis Group) in Physical Geography and Environment, Department of Geography and Environment, University of Dhaka
Thesis: Understanding the effects of Urban Morphological Indicators on Land Surface Temperature in Dhaka North City Corporation: Combining Machine Learning and MGWR models

Research Grant

January 2026 Government's Research Grant (2025-26), Department of Disaster Management, Ministry of Disaster Management and Relief, People's Republic of Bangladesh
Research Title: Assessing Community Earthquake Preparedness, Perception and Response in Madhabdi and Narsingdi Municipality, Narsingdi

Research Experience

November 2025 – December 2025

Project Research Assistant

Carbon Footprint Mapping of Humanitarian Operations in Rohingya Refugee Camps funded by **UNHCR**, implemented by **NGO Forum for Energy**

Notable Activities: Literature Review, five working days-long Fieldwork in the Rohingya Refugee Camp in Ukhia and Teknaf in Cox's Bazar, Preparing FGD Questionnaires, Drafting Data Requisition Checklists, collecting data, data processing in Excel, visualization and mapping in ArcGIS desktop, Report Writing and Documentation.

December 2024 – May 2025

Research Assistant

GIS-based Fire Hazard Vulnerability Mapping of Residential turned Eatery buildings in Dhaka City funded by **DRTMC, University of Dhaka**.

Notable Activities: Literature Review, Frequent Field Visits, Preparing Questionnaires, collecting data via Kobo Collect, data processing in Excel, visualization and mapping in ArcGIS desktop, Report Writing, Manuscript Writing for Journal Article.

October 2023 – April 2024

Project Research Assistant, Project on Situation Analysis of Fire Disaster in Dhaka City, conducted and funded by **Disaster Research Training and Management Centre (DRTMC), University of Dhaka (2023-24)**.

Notable Activities: - As Project Research Assistant, I was assigned the task of literature review, questionnaire preparation, data collection, data analysis and report writing.

- Co-author of the report submitted to DRTMC on "Exploring the long sustenance of Fire Hazard in Dhaka city"

Fieldwork Experience

<u>February 2025</u>	Land Use and Land Cover Survey and Socio-economic dynamics of Chunarughat Mouza, Habiganj. Activities included: georeferencing and digitizing mouza map, preparing questionnaire and survey, GIS-based mouza mapping, RS-based LULC Change Detection using ArcGIS Desktop, SPSS and Excel for Data analysis and visualization.
<u>February 2024</u>	Hydro-morpho-ecological study of Dinajpur District. Activities included: surveying river flow, vegetation, soil monolith, ecology of Ashurar Beel, usage of GPS meter, Sediments Grabber, analyzing data using Excel and Mapping Using ArcGIS Desktop Software
<u>June 2023</u>	Socio-economic Survey of Asam Basti Community, Rangamati. Activities included: preparing questionnaires, surveying people from different ethnic background, Data analysis using SPSS and Excel, making maps using ArcGIS desktop.
<u>September 2022</u>	Surveying the Problem and Prospects of Shrimp Cultivation in Shymnagar Upazila. Satkhira and Vegetation of Sunderbans Mangrove Forest. First-hand experience with salinity intrusion in the coastal Bangladesh and people's adaptation to it.

Computer Literacy

<u>Program</u>	<u>Level of Expertise</u>	<u>Program</u>	<u>Level of Expertise</u>
Microsoft Office	Advanced User	ArcGIS Desktop	Intermediate
ArcGIS Pro	Intermediate	GEE	Intermediate
SPSS	Intermediate	Kobo Collect	Intermediate
Canva	Advanced	Adobe Illustrator	Intermediate
Python	(NumPy, Pandas, Matplotlib, Seaborn, Geopandas, Rasterio)	R	(Tidyverse, Leaflet, ggplot2)

Awards and Recognitions

First Runner Up	International Day for Disaster Risk Reduction Poster Presentation Competition. Poster on “Climate Resilience of Coastal Communities in Southwestern Bangladesh and How Fighting Inequality can bolster their endeavor”
Final Round (6 th)	National Policy Competition 2025, Dhaka University Chapter organized by Ministry of Youth and Sports, People's Republic of Bangladesh.
Certification	Training Workshop on Environmental Management Systems (ISO 14001:2015), Issued by AJA Bangladesh

Language Proficiency

Bangla (Native)	English (Fluent)
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Writing In Newspapers

Khandaker, H. (November 20, 2023) Rethinking University Education, The Financial Express. Available at: <https://today.thefinancialexpress.com.bd/editorial/rethinking-university-education-1700402178> (Accessed: 14 January 2025).

Khandaker, H. (July 04, 2024) Our teachers deserve better, The Financial Express. Available at: <https://today.thefinancialexpress.com.bd/editorial/our-teachers-deserve-better-1720015051> (Accessed: 14 January 2025).